

DESIGN OF A 3 BIT ALU USING VERILOG IN VIVADO

AIM

To design, simulate, synthesize, and verify an **Arithmetic Logic Unit (ALU)** supporting **8 operations** using **Xilinx Vivado**.

DESIGN SPECIFICATION

INPUTS

A - 3 Bit Operand

B - 3 Bit Operand

Sel - 3 Bit Selectline to choose the operation

OUTPUT

Y - 3 Bit Output

Carry - 1 bit Output

OPERATIONS

Select Line	Operation Name	In Symbol
000	ADDITION	$A + B$
001	SUBTRACTION	$A - B$
010	AND	$A \& B$
011	OR	$A B$
100	XOR	$A \wedge B$
101	NOT	$\sim A$
110	LEFT SHIFT	$A \ll 1$
111	RIGHT SHIFT	$A \gg 1$

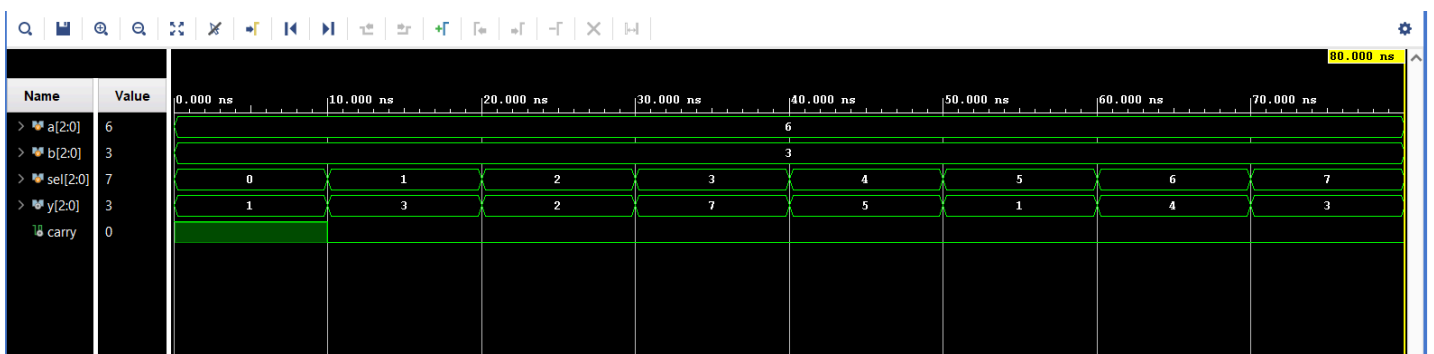
SOURCE CODE

```
module alu_3bit (input [2:0] a,input [2:0] b,input [2:0] sel,output reg [2:0] y,output reg carry);
always @(*) begin
    carry = 1'b0;
    case (sel)
        3'b000: {carry, y} = a + b; //Addition
        3'b001: {carry, y} = a - b; //Subtraction
        3'b010: y = a & b; //AND
        3'b011: y = a | b; //OR
        3'b100: y = a ^ b; //XOR
        3'b101: y = ~a; //NOT
        3'b110: y = a << 1; //LEFT SHIFT
        3'b111: y = a >> 1; //RIGHT SHIFT
        default: y = 3'b000;
    endcase
end
endmodule
```

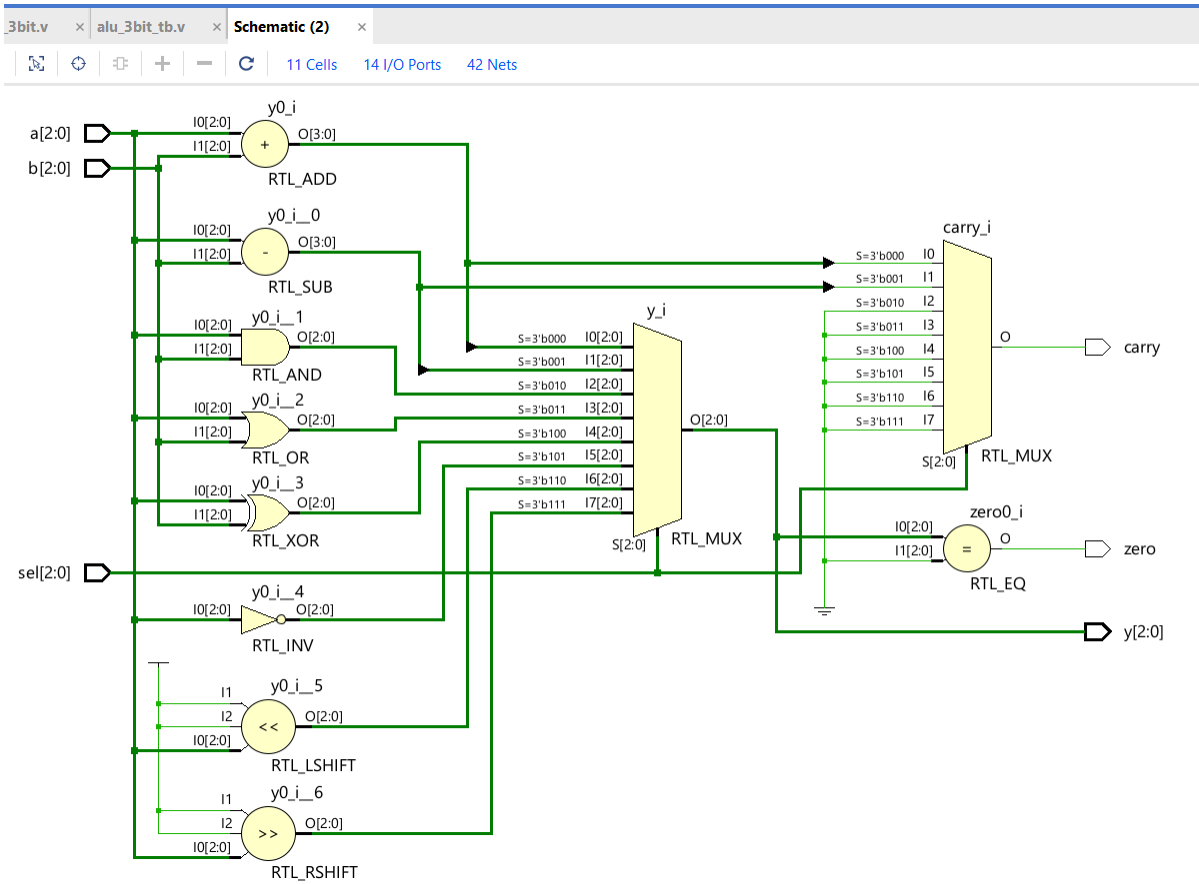
TESTBENCH

```
30
31     alu_3bit uut (a,b,sel,y,carry
32 );
33
34 initial begin
35
36     a = 3'b110; b = 3'b011;
37
38     sel = 3'b000; #10;
39     sel = 3'b001; #10;
40     sel = 3'b010; #10;
41     sel = 3'b011; #10;
42     sel = 3'b100; #10;
43     sel = 3'b101; #10;
44     sel = 3'b110; #10;
45     sel = 3'b111; #10;
46
47     $finish;
48 end
49 endmodule
```

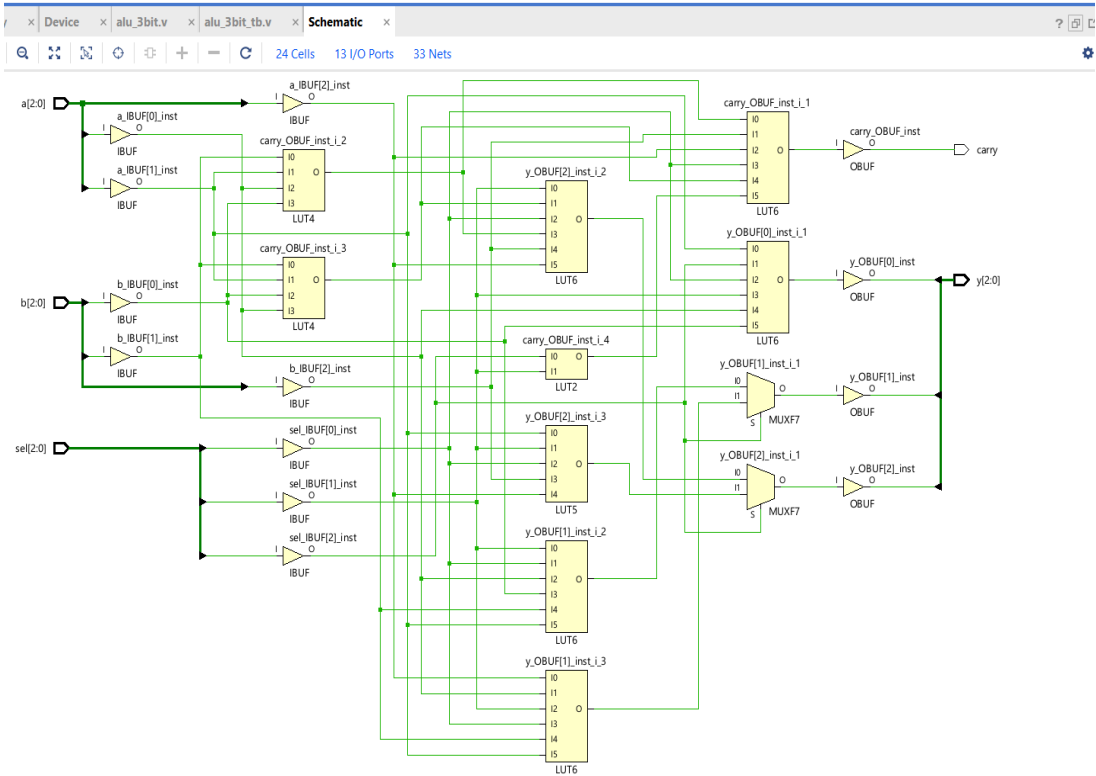
SIMULATION OUTPUT



RTL SCHEMATIC



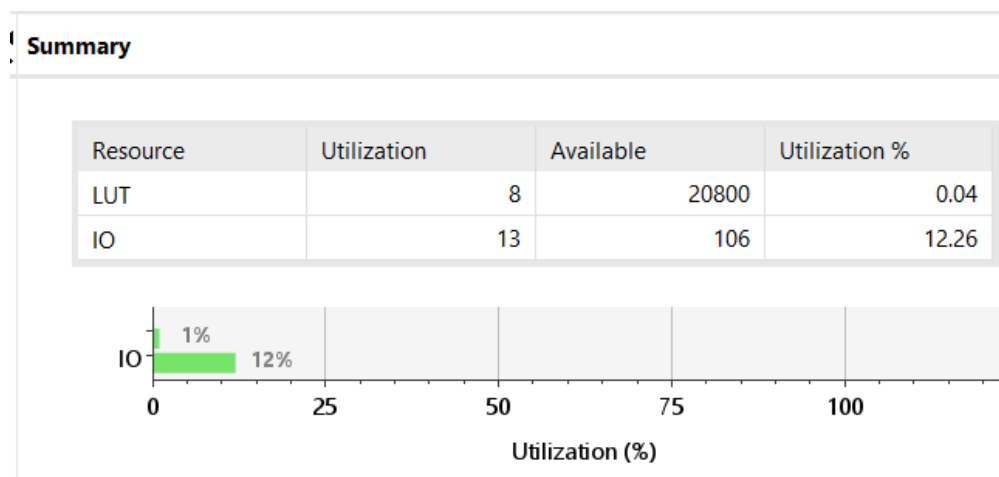
SYNTHESIZED SCHEMATIC



WORKING PROCEDURE

1. Read input operands A and B
2. Read select line Sel
3. Based on Sel, choose one of the 8 operations
4. Perform the operation using case(sel)
5. Store result in output Y
6. Set carry for arithmetic operation where applicable
7. Display results in simulation waveform

Synthesis Resource Utilization



Observed resource usage

- LUTs used: 8
- I/O ports used: 13
- F7 multiplexers used: 2

Conclusion

The 3-bit ALU was successfully designed and implemented using **Verilog HDL in Xilinx Vivado**. It performs **8 arithmetic and logical operations** such as addition, subtraction, AND, OR, XOR, NOT, left shift, and right shift based on the select input. The design was verified through simulation, and the results confirmed correct operation for all cases. The synthesis report also showed that the ALU uses very few FPGA resources, making it a simple and efficient digital design.